Information-Triggered Delivery for Timely Victim Reporting in Infrastructure-Free UAV Search and Rescue

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***Abstract*—Small UAV swarms with onboard thermal detectors can search disaster areas without infrastructure, but a victim found by a UAV helps only once the report reaches the responders, over radio links that are obstructed, fading and broken by UAV failures. Keeping the swarm connected at all times makes every report immediate but confines the search; searching freely leaves delivery to chance. We present SENTINEL, a decentralised framework that lets UAVs search freely and triggers delivery from the information each UAV has confirmed, fuses thermal detections from several viewpoints so that occluded angles cannot veto a clear one, and, while a report is being delivered, constrains the UAVs on the delivery path with an outage model and a local minimum spanning tree so that preserved links remain reliable for every feasible step. We bound the probability that a confirmed detection is lost before it is secured and show that nearest-neighbour distance constraints do not preserve connectivity. In about 13 000 simulated missions against seven baselines tuned with the same budget (connectivity control, potential fields, communication-aware PSO, MAPPO and sweep search), with 30 seeds per setting and paired tests, SENTINEL reports 23–30 % more victims within 60 s than the best baseline in urban rubble in six failure settings, is significantly better than all baselines for 3–50 UAVs on a 400 m area, and is never significantly worse than any of them; PX4 software-in-the-loop flights in Gazebo agree with the fast simulator. Ablations show significant effects of the fusion rule and of the delivery trigger. Given the same fusion rule, the baselines remain significantly behind in the urban 400 m scenarios, but match or exceed SENTINEL in a dense 50-UAV swarm, where fusion alone explains its advantage. Continuous connectivity does not help. The onboard YOLO26n detector matches YOLOv8n in validation accuracy on HIT-UAV with 31 % less compute.**

***Index Terms*—UAV swarms, intermittent connectivity, search and rescue, edge AI, particle swarm optimization, delay-tolerant networking.**

I. Introduction

After earthquakes, floods and large fires, cellular and backhaul infrastructure is often unavailable precisely where responders need information. Small multirotor UAVs carrying thermal cameras can search such areas quickly, and lightweight detectors trained on aerial thermal imagery [1] need only a few GFLOPs per frame, so victims can be detected on board without streaming video to a server. What remains hard is getting each detection to the responders *in time*. Without infrastructure, a report travels only over the UAVs' own short-range radio links, which are obstructed by buildings, fade, and break when UAVs fail.

Two families of methods address the communication side. Connectivity-maintenance controllers keep the swarm's communication graph connected at all times, either through the algebraic connectivity of the graph [2], [3] or through edge potentials that never let a link break [4]. Every detection then reaches the responder immediately, but the whole swarm is tied to the responder's neighbourhood, which limits how far it can search. Intermittent-connectivity and data-ferrying approaches let robots disconnect and meet according to a schedule [5], or plan ferry routes to sources whose positions are known in advance [6], [7]. In search and rescue neither assumption fits well: the information sources, the victims, are *discovered by the search itself*, at unknown places and times.

In an earlier version of this system we coupled a thermal detector with particle swarm optimisation (PSO) and a distance penalty that kept UAVs within 50 m of their nearest neighbour. That design has three limitations, which also shaped this work. A fixed distance does not describe a radio link behind rubble; keeping the nearest neighbour close does not keep the network connected (Lemma 1); and the evaluation used a 25 m area that is far smaller than the radio range, in which, as we show in Section VI-J, every method stays connected and the methods differ by at most eight percentage points.

This paper studies the *discovery–delivery trade-off* directly. We measure the share of victims whose report reaches the responder within a deadline, and we treat connectivity as a means to that end rather than as an objective. Our contributions are:

- **Information-triggered delivery.** Each UAV searches without connectivity constraints and switches to a delivery mode when a victim it confirmed has not reached the responder after a hold time T_h (Section IV). Rendezvous are triggered by information that the search itself discovers, rather than by a schedule or by known source locations. T_h exposes the trade-off explicitly, and we bound the probability that a confirmed detection is lost to UAV failures before it is secured (Proposition 3).
- **Corroborative multi-view sensing.** Weak detections request a lateral second view from a neighbour, and views from different directions are fused so that occluded viewpoints cannot veto a clear one (Section IV-G). In dense swarms this fusion rule is the largest single contributor to performance.
- **A link-preservation safety layer with guarantees.** While a delivery is in progress, the UAVs on the delivery path constrain their steps using a realistic outage model (path loss, obstructions, shadowing, fading, interference) and a local minimum spanning tree. Preserved links remain reliable for every feasible step (Proposition 1), and

the delivering component stays connected (Corollary 1). We show why nearest-neighbour constraints do not suffice (Lemma 1).

- **A reproducible evaluation.** We compare against seven baselines covering communication-aware optimisation, algebraic-connectivity control, potential-based topology control, multi-agent reinforcement learning and classical sweep search, all tuned with the same budget. The evaluation spans 3–50 UAVs, four channel profiles, five failure models and a failure-free reference, moving obstacles and detector uncertainty, uses 30 seeds per configuration with paired significance tests, and is cross-checked against PX4 software-in-the-loop flight in Gazebo. Code, scenarios and results are released.

Across the scenarios SENTINEL reports 23–30 % more victims within 60 s than the best baseline in urban rubble in all failure settings, and it is significantly better than all baselines on the 400 m area at every swarm size from 3 to 50; in the harshest channel it is ahead of or tied with the best baseline. The ablation shows which components matter: the fusion rule and the delivery trigger have significant effects; given the same fusion rule, the baselines remain significantly behind in the urban 400 m scenarios. The link-preservation layer has no significant effect on the on-time share but reduces preserved-link violations 37-fold compared with the same tethers without it. We also report what does not work as well: keeping the swarm connected at all times lowers on-time reporting to 0.03–0.13; in small, dense swarms a MAPPO policy trained to convergence comes close to SENTINEL (differences not significant for $N = 10$–20 on the 200 m area); in a dense 50-UAV swarm, baselines given SENTINEL's fusion rule match it and one exceeds it; and the thermal detector, trained on real imagery, transfers poorly to rendered Gazebo scenes.

Section II positions this work against prior approaches, Section III defines the system model and problem, Section IV presents SENTINEL, Section V describes the implementation, Section VI reports the evaluation and Section VII discusses limitations.

II. Related Work

a) Continuous connectivity maintenance: A large body of work keeps a multi-robot communication graph connected at all times. Zavlanos and Pappas control the robots so that the graph Laplacian keeps a positive second eigenvalue [3]; Sabattini et al. estimate the algebraic connectivity λ_2 in a decentralised way and follow its gradient without requiring local links to persist [2]; Ji and Egerstedt preserve edges through potentials that grow unbounded as a link approaches its range, with hysteresis on edge creation [4]. Local topology-control rules such as the local minimum spanning tree (LMST) [8] reduce the set of links that must be kept. These methods prove that connectivity is maintained, but under a disk communication model and at the price of tying every robot to the network at every instant. Our baselines B3 and B4 implement the second and third approach (B3 with the exact λ_2), and our measurements show that in large areas this tie lowers the share of victims reported in time. SENTINEL applies such constraints only for the duration of a delivery (Proposition 1) and replaces the disk model by a predicted outage model that accounts for obstructions and fading.

b) Intermittent connectivity, data ferrying and information age: Kantaros and Zavlanos let robots disconnect and enforce that they meet at the vertices of a mobility graph infinitely often, with meeting schedules specified in temporal logic [5]. UAV data-ferrying and delay-tolerant designs carry data physically between sources and a sink [6], [9], and age-of-information (AoI) formulations optimise how fresh the collected data are [7], [10], [11]. These approaches assume that the sources (sensors) are known in advance or that meetings follow a predefined schedule. In search and rescue the sources are the victims, discovered online by the search itself; SENTINEL's rendezvous are triggered by that discovered information, and our deadline metric is AoI-like (time from confirmation to the responder).

c) Communication-aware swarm search and learning: PSO and its variants are widely used for multi-UAV coverage and search [12], [13]; distributed planners such as DAPSO run a local PSO on each UAV and exchange information over local links, but assume that communication is available when needed [14]. Communication-aware variants add link-quality terms to the fitness; our baseline B2 represents this family with the same outage model as SENTINEL applied as a soft penalty to all neighbours. Distributed Bayesian search reduces traffic by flooding only positive detections but assumes a connected, lossless network [15], and decentralised mountain-SAR search uses distance-limited links without a responder sink [16]. Multi-agent reinforcement learning, with MAPPO as a strong general baseline [17], has been applied to UAV coverage and communication-aware coordination [18]; we include MAPPO with the same observations and objective as baseline B5.

d) Onboard thermal perception: Thermal datasets such as HIT-UAV [1] enable lightweight detectors that run on the UAV, and recent YOLO generations target edge deployment by removing non-maximum suppression and reducing compute [19]. We use such a detector as an enabling component and evaluate it on real thermal imagery; we do not claim it as a contribution.

e) Positioning: Table I summarises the comparison. To our knowledge, no prior work combines online-discovered information sources, an explicit reporting deadline, a channel model with obstructions and fading, failure-aware evaluation, and provable link preservation during delivery (Proposition 1).

III. System Model and Problem Formulation

A. Network and Mobility

A swarm $\mathcal{U} = \{1, \dots, N\}$ of multirotor UAVs flies at a common altitude h over a disaster area $\mathcal{A} \subset \mathbb{R}^2$ containing static obstacles. Obstacles taller than h form a no-fly set $\mathcal{O} \subset \mathcal{A}$ and obstruct radio links; low rubble only occludes ground targets. A responder radio (node 0) is fixed at the launch point $x_0 \in \mathcal{A}$; it is the only sink for victim reports and is assumed not to fail. UAV i has position $x_i(t) \in \mathcal{A} \setminus \mathcal{O}$ and speed $\|\dot{x}_i\| \le v$. Time is slotted with control period Δt; every UAV broadcasts one beacon per slot containing its position, its target ledger and acknowledgements (Section IV).

TABLE I
COMPARISON WITH THE CLOSEST APPROACHES (✓: ADDRESSED; ○: PARTIALLY; –: NOT ADDRESSED).

Approach	Sources discovered online	Delivery deadline / timeliness metric	Obstruction/fading channel model	UAV failures	Connectivity proof (Prop. 1)	Free search (no permanent tie)
λ_2 control [2], [3]	–	–	–	○	✓	–
Edge potentials [4]	–	–	–	–	✓	–
Intermittent connectivity [5]	–	○	–	–	✓	✓
Data ferrying / DTN [6], [9]	–	✓	○	–	–	✓
AoI UAV data collection [7], [10], [11]	–	✓	○	–	○	✓
Distributed PSO search [13], [14]	✓	–	–	–	–	✓
Event-driven Bayesian search [15]	✓	–	–	–	–	✓
MARL coordination [17], [18]	○	○	○	○	–	✓
Earlier SENTINEL design (B1)	✓	–	–	○	–	○
SENTINEL (this work)	✓	✓	✓	✓	✓†	✓

† during delivery, for feasible steps (Proposition 1, Corollary 1).

TABLE II
CHANNEL PROFILES ($f = 2.4$ GHz, $P_t = 20$ dBm, $G = 2$ dBi, $B = 20$ MHz, NOISE FIGURE 7 dB, $\gamma_{\rm th} = 5$ dB) AND RESULTING RELIABLE RANGES AT $\varepsilon = 0.05$ (σ_X IN dB).

Profile	η	σ_X	K (LoS)	L_o	$R(0)$	$R(1)$
open	2.0	2	10 dB	6 dB	2434 m	684 m
suburban	2.5	4	6 dB	10 dB	339 m	95 m
urban-rubble	3.0	6	3 dB	6 dB	91 m	48 m
severe	3.5	8	Rayleigh	8 dB	35 m	21 m

B. Channel Model

The received SINR of the broadcast from i at j in slot t is

$$\gamma_{ij}(t) = \frac{P_t G^2\, 10^{-(\mathrm{PL}(d_{ij}) + n_{ij} L_o + X_{ij})/10}\, |h_{ij}|^2}{\sigma^2 + \sum_{k \in \mathcal{I}_j(t)} \bar{P}_{kj}}, \qquad (1)$$

where P_t is the transmit power, G the antenna gain per side, σ^2 the noise power, $\mathrm{PL}(d) = \mathrm{PL}(d_0) + 10\eta \log_{10}(d/d_0)$ log-distance path loss (in dB, as are L_o and X_{ij}), n_{ij} the number of obstacles crossing the segment $[x_i, x_j]$ with per-obstruction loss L_o, $X_{ij} \sim \mathcal{N}(0, \sigma_X^2)$ (dB) log-normal shadowing evolving with the Gudmundson correlation $\rho = e^{-\Delta/d_c}$ for endpoint displacement Δ, $|h_{ij}|^2$ unit-mean Rician fading with factor K ($K = 0$, Rayleigh, once obstructed), and $\mathcal{I}_j(t)$ the set of transmitters colliding in the slot, each present with probability p_a. A packet is decoded iff $\gamma_{ij} \geq \gamma_{\rm th}$. The link outage probability, averaged over fading and shadowing, is

$$P_{\rm out}(d, n) = \mathbb{E}_X \left[F_{|h|^2}\!\left(\frac{\gamma_{\rm th}}{\bar{\gamma}(d, n)\, 10^{X/10}} \right) \right], \qquad (2)$$

with $F_{|h|^2}$ the non-central χ^2_2 CDF (computed in closed form; the outer expectation by Gauss–Hermite quadrature). The *reliable range* for target outage ε is $R_\varepsilon(n) = \max\{d : P_{\rm out}(d, n) \leq \varepsilon\}$, and the *reliable graph* $\mathcal{G}_\varepsilon(t)$ joins i, j iff $P_{\rm out}(d_{ij}, n_{ij}) \leq \varepsilon$. Table II lists the four environment profiles; their parameters follow the ranges reported for low-altitude aerial and street-level links [20], [21], with shadowing correlation following [22]; they are not calibrated against measurements in rubble (Section VII). For the urban-rubble profile $R_{0.05}(0) = 91$ m and $R_{0.05}(1) = 48$ m, which places the fixed 50 m range of the earlier SENTINEL design in context: it is the reliable range of a link behind one obstruction, not a property of the radio alone.

C. Sensing Model

Each UAV carries a nadir thermal camera with ground footprint radius r_f and an onboard detector (Section V). A look at a cell updates the residual probability that an undetected victim is present by Bayes' rule, $p \leftarrow p(1 - P_d)/(1 - pP_d)$ [23]. For a victim at slant range r the detector returns a confidence $c \sim \mathrm{clip}_{[0,1]} \mathcal{N}(\mu, \sigma_c^2)$ with $\mu = c_{\max}(1 - o)^\gamma e^{-[r - r_{\rm ref}]_+ / r_s}$, where the occlusion $o = \mathrm{clip}_{[0,1]}\big(o_0 + \kappa_o \sin\vartheta \cos(\phi - \phi_b)\big)$ grows when the victim is viewed from the side of the covering structure (off-nadir angle ϑ, azimuth ϕ, blocking direction ϕ_b, base occlusion o_0; $o = 1$ for buried victims). False positives from warm debris arrive as a Poisson process per look with low confidences. The parameters (Table III) are set to reproduce the qualitative behaviour of the onboard detector, not fitted to it, and Section VI-G sweeps them.

D. Information Model

Every UAV keeps a ledger of world-frame target entries. Observations within a gate are associated to one entry; the best confidence per viewing sector is kept and the sectors are fused into one confidence c_e (Section IV-G). An entry is *confirmed* when $c_e \geq c_{\rm hi}$ and *tentative* when $c_{\rm lo} \leq c_e < c_{\rm hi}$. Ledgers are replicated by gossip with acknowledgements, so each entry carries its holder set $H_e(t)$. An entry is *secured* once $0 \in H_e$ (the responder holds it) and *lost* if it was confirmed but no alive node holds it.

E. Failure Model

UAVs fail by crashing (the node and its ledger copy disappear), by radio loss (it flies but cannot communicate) or by gradual brownout (transmit power decays). We study single, simultaneous, cascading and Poisson failures with per-UAV rate λ_f.

F. Problem

Let τ_k be the time victim k is first confirmed by any UAV and $T_k \geq \tau_k$ the time, measured from the start of the mission, at which its report reaches the responder ($T_k = \infty$ if this does not happen before the mission horizon T). With value discount θ, the team maximises the time-discounted report value

$$J = \frac{1}{|\mathcal{V}|} \sum_{k \in \mathcal{V}} e^{-T_k/\theta}, \qquad (3)$$

subject to the dynamics, $x_i \notin \mathcal{O}$, and decentralised information (each UAV acts on its own sensing and on beacons it actually received). J couples three quantities that prior work optimises separately: discovery (a victim must be found), delivery (its report must reach the responder), and timeliness (sooner is better, in the spirit of age of information [10]). Maintaining $\mathcal{G}_\varepsilon$ connected at all times makes delivery immediate but restricts discovery; pure exploration maximises discovery but leaves delivery to chance contacts, so confirmed information can wait arbitrarily long or be lost with its carrier. We refer to this tension as the *discovery–delivery trade-off*.

IV. SENTINEL

Each UAV runs the same decentralised agent (Algorithm 1), built from three parts. (i) *Information-triggered delivery*: the agent alternates between two modes driven by its own information state; in EXPLORE it searches without any connectivity constraint, in DELIVER it returns toward the responder until the victims it has confirmed are secured. (ii) *Corroborative multi-view sensing*: weak detections request a second viewpoint from a neighbour, and views are fused so that occluded angles cannot veto a clear one. (iii) *A link-preservation safety layer*: while delivering, a UAV and the relays between it and the responder constrain their motion so that the links they rely on stay reliable, with a guarantee for feasible steps (Proposition 1). Section VI-H shows that (i) and (ii) are the components with significant effects on performance, while (iii) keeps deliveries from being interrupted by the swarm's own motion.

A. Mode switching

Let $\mathcal{E}_i(t)$ be the confirmed entries held by UAV i that are not secured and for which i is responsible: it created them, or they are older than $3T_h$, so that a copy whose creator failed is eventually delivered by another holder. UAV i switches to DELIVER when the oldest such entry reaches age T_h,

$$\max_{e\in\mathcal{E}_i(t)} (t-\tau_e) \;\geq\; T_h, \tag{4}$$

and back to EXPLORE when $\mathcal{E}_i(t)=\emptyset$. The hold time T_h is the single parameter that trades discovery against delivery latency: $T_h = 0$ delivers every confirmation immediately, $T_h \to \infty$ never interrupts the search. Whether a UAV is connected to the responder is known locally from a heartbeat: the responder stamps its beacons with the current time and every UAV relays the freshest stamp it received; UAV i is *connected* if its freshest stamp is younger than T_{hb}.

B. Waypoint optimisation (EM-PSO)

Every planning period the agent chooses its next waypoint inside the reachable disk $\mathcal{D}_i = \{x : \|x - x_i\| \leq vT_p\} \setminus \mathcal{O}$ by maximising

$$F_i(x) = w_c\,\hat{G}_i(x) + w_d \sum_{e\in\mathcal{T}_i} c_e\, e^{-\frac{\|x-x_e\|^2}{2s^2}} + \frac{w_g(\|x_i-g_i\| - \|x-g_i\|) - w_t\|x-x_i\|}{vT_p} - S_i(x), \tag{5}$$

where $\hat{G}_i(x)$ is the normalised expected number of detections of a look at x computed on the UAV's own search map (cells inside footprints claimed by neighbours removed), $\mathcal{T}_i$ the tentative entries, g_i a frontier goal and w_t a small travel cost. The frontier goal is the map cell z maximising $\hat{G}_i(z)\,(1 + b_s \mathbb{1}[z \in \Theta_i])/(1 + \|z - x_i\|/\ell_g)$, where Θ_i is an angular sector around the responder assigned to i by its identifier and cells near neighbours' waypoints are excluded; w_g and b_s are tuned exploration aids and not claimed as contributions. The *signal spring* acts only while the UAV is tethered (in DELIVER and connected, or holding a relay tether, Section IV-D):

$$S_i(x) = \lambda_i \sum_{j\in\mathcal{S}_i} \left(\frac{[P_{\mathrm{out}}(\|x-x_j\|, n(x,x_j)) - \varepsilon_i]_+}{\varepsilon_i} \right)^2, \tag{6}$$

with $\lambda_i = \lambda_0(1+\beta U_i)$ and $\varepsilon_i = \varepsilon/(1+\beta U_i)$, where U_i is the number of entries only i holds. Unlike a distance penalty, (6) penalises the predicted *outage* of the link, so it accounts for obstructions and fading, and it tightens with the amount of unique information the UAV carries. The maximisation uses a constriction-coefficient PSO ($\chi = 0.7298$, $c_1 = c_2 = 2.05$) [12], whose particles are seeded with the previous waypoint, the step toward g_i and the waypoints broadcast by neighbours.

C. Link preservation (safety layer)

The spring (6) steers the search toward reliable links but cannot ensure that they survive. While tethered, UAV i therefore projects its step onto a feasible set built from its preserved neighbours $\mathcal{S}_i$.

Definition 1 (Local MST neighbours)**.** Let $\hat{\mathcal{G}}_i$ be the subgraph of the reliable graph induced by i and the neighbours it heard within the last T_{nb}, with edge weights $(d_{jk}, \max(j,k), \min(j,k))$ ordered lexicographically. $\mathcal{S}_i$ is the set of neighbours adjacent to i in the minimum spanning tree of $\hat{\mathcal{G}}_i$ [8].

Lemma 1 (Nearest-neighbour constraints do not imply connectivity)**.** *There are configurations in which every UAV is within R of its nearest neighbour while the communication graph is disconnected.*

Proof. Four UAVs at $(0,0), (10,0), (200,0), (210,0)$ with $R = 50\,\mathrm{m}$: each has a neighbour at 10 m, yet no edge joins $\{1,2\}$ to $\{3,4\}$. □

Lemma 1 is why the nearest-neighbour spring of the earlier SENTINEL design cannot certify network maintenance.

Lemma 2 (LMST spans the reliable graph [8])**.** *Let $\mathcal{G}^-$ contain edge (i,j) iff $j \in \mathcal{S}_i$ and $i \in \mathcal{S}_j$, all computed from the same positions, and assume every UAV has heard each of its reliable neighbours within T_{nb}. Then $\mathcal{G}^-$ contains the minimum spanning forest of the reliable graph, hence has the same connected components.*

Proof. Let (i,j) be an edge of the global minimum spanning forest $\mathcal{M}$. If (i,j) were not in the local MST of $\hat{\mathcal{G}}_i$, the cycle property would give a cycle in $\hat{\mathcal{G}}_i$ on which (i,j) is the unique heaviest edge (by the assumption, $\hat{\mathcal{G}}_i$ contains all reliable edges

at i, and its other edges are reliable); that cycle exists in the reliable graph, contradicting $(i,j) \in \mathcal{M}$. The same holds for $\hat{\mathcal{G}}_j$, so $(i,j) \in \mathcal{G}^-$. Since $\mathcal{M} \subseteq \mathcal{G}^-$ and $\mathcal{G}^-$ only uses reliable links, the components coincide. □

a) Midpoint-ball projection: For each $j \in \mathcal{S}_i$ let $\hat{x}_j$ be j's last reported position, of age a_{ij}, $m_{ij} = (x_i + \hat{x}_j)/2$, and $\delta_{ij} = v(\Delta t + a_{ij})$ the furthest j can move before the next check. The proposed step u_i is scaled to the largest $\alpha \in [0,1]$ such that $x_i' = x_i + \alpha u_i$ satisfies, for all $j \in \mathcal{S}_i$,

$$\|x_i' - m_{ij}\| \le \tfrac{1}{2} r_{ij}(x_i') - \tfrac{1}{2} v\, a_{ij}, \quad r_{ij}(x_i') = R_\varepsilon\big(n^{\delta_{ij}}(x_i', \hat{x}_j)\big), \tag{7}$$

where $n^\delta(a,b)$ counts obstacles inflated by δ that cross $[a,b]$. If no $\alpha > 0$ satisfies (7) the UAV hovers, or, if it is already outside a constraint, takes the largest step toward the violated centre that does not enlarge any other violation. Each UAV evaluates (7) with local information only.

Proposition 1 (Link preservation)**.** *Suppose $j \in \mathcal{S}_i$ and $i \in \mathcal{S}_j$, both UAVs are tethered, speeds are bounded by v, positions are exchanged with staleness at most a, and the steps of both UAVs satisfy* (7)*. Then the link (i,j) is ε-reliable at $t + \Delta t$.*

Proof. Let $m = (x_i + x_j)/2$ be the true midpoint. Since $\|\hat{x}_j - x_j\| \le v a_{ij}$, we have $\|m_{ij} - m\| \le v a_{ij}/2$ and, symmetrically, $\|m_{ji} - m\| \le v a_{ji}/2$. By (7) and the triangle inequality, $\|x_i' - x_j'\| \le \|x_i' - m_{ij}\| + \|m_{ij} - m\| + \|m - m_{ji}\| + \|m_{ji} - x_j'\| \le \frac{1}{2}(r_{ij} + r_{ji})$. The segment $[x_i', x_j']$ lies within $\|x_j' - \hat{x}_j\| \le \delta_{ij}$ of $[x_i', \hat{x}_j]$, so every obstacle crossing it, un-inflated, crosses $[x_i', \hat{x}_j]$ when inflated by δ_{ij}: $n(x_i', x_j') \le n^{\delta_{ij}}(x_i', \hat{x}_j)$ and, R_ε being non-increasing in n, $R_\varepsilon(n(x_i', x_j')) \ge r_{ij}$. Likewise $R_\varepsilon(n(x_i', x_j')) \ge r_{ji}$. Hence $\|x_i' - x_j'\| \le \max(r_{ij}, r_{ji}) \le R_\varepsilon(n(x_i', x_j'))$. □

Corollary 1 (Connectivity during delivery)**.** *If all UAVs of a connected component of $\mathcal{G}_\varepsilon(t)$ containing the responder are tethered, take steps satisfying* (7)*, and no UAV fails, that component remains connected at $t + \Delta t$.*

Proof. By Lemma 2 the mutually selected edges span the component, and by Proposition 1 each of them is reliable at $t + \Delta t$ (the responder never moves, so a one-sided constraint suffices for its edges). □

Remark. Proposition 1 is a one-step guarantee for feasible steps; it does not by itself make (7) feasible at every step (the inflated obstruction count can shrink r_{ij} even for a hovering UAV). We therefore also report the empirical rate of preserved-link violations in all experiments, and a Monte-Carlo check of the obstacle-free synchronous case with exact positions (10^4 random swarms of 2–50 UAVs, 25 adversarial steps each, no violation).

Proposition 2 (Probabilistic delivery path)**.** *Under Corollary 1, the probability that, in a slot, the realised (fading) network restricted to the preserved spanning tree $\mathcal{M}$ is disconnected satisfies* $\Pr[\textit{disc.}] \le \sum_{e \in \mathcal{M}} P_{\mathrm{out}}(e) \le (|\mathcal{M}|)\,\varepsilon$.

Proof. The realised tree is disconnected only if at least one of its edges is in outage; apply the union bound and $P_{\mathrm{out}}(e) \le \varepsilon$ for preserved edges. □

D. Relay tether

Proposition 1 needs *both* endpoints tethered, but the relays between a delivering UAV and the responder are usually exploring. A delivering, connected UAV therefore stamps $t_{\mathrm{tether}} = t + T_{\mathrm{rel}}$ in its beacon; stamps propagate by maximum, and a connected UAV is tethered while $t \le t_{\mathrm{tether}}$. The whole connected component thus holds its links during a delivery and is released T_{rel} after it ends; because the stamp is set only by delivering UAVs, tethers cannot sustain each other indefinitely.

E. Information exposure

Proposition 3 (Exposure bound)**.** *Suppose UAVs fly at speed v along paths at most $\kappa \ge 1$ times the straight-line distance, a delivering UAV completes the hand-off within τ_c of reaching the responder's connected component, and an entry confirmed at τ_e is held from τ_e on by k UAVs (its creator and $k-1$ replicas). Let $D = \max_{x \in \mathcal{A}} \|x - x_0\|$, $W = T_h + \kappa D/v + \tau_c$ and $\bar{W} = 3T_h + \kappa D/v + \tau_c$ $(\ge W)$. If the holders fail independently at rate λ_f,*

$$\Pr[\textit{entry lost}] \le \left(1 - e^{-\lambda_f W}\right)\left(1 - e^{-\lambda_f \bar{W}}\right)^{k-1} \le (\lambda_f \bar{W})^k. \tag{8}$$

Proof. If the creator is alive at $\tau_e + W$, the entry is secured by then: by (4) DELIVER starts at most T_h after confirmation, the return takes at most $\kappa D/v$ and the hand-off at most τ_c, after which $0 \in H_e$ (the responder does not fail). Otherwise the creator fails before $\tau_e + W$. Every other holder becomes responsible for the entry when it is older than $3T_h$, i.e. at the latest at $\tau_e + 3T_h$; the entry is then older than T_h, so a surviving holder switches to DELIVER at once and secures the entry by $\tau_e + 3T_h + \kappa D/v + \tau_c = \tau_e + \bar{W}$. Hence the entry is lost only if the creator fails within W and each of the other $k-1$ holders fails within $\bar{W}$, which under independence has the stated probability; the last step uses $1 - e^{-z} \le z$ and $W \le \bar{W}$. □

Proposition 3 makes the role of T_h explicit: it bounds the exposure of confirmed information to UAV failures (the information-loss side), while larger T_h leaves more time for search (the discovery side). Section VI measures the resulting trade-off and compares the measured losses with (8).

F. Collaborative synthetic-aperture sensing

A tentative entry ($c_{\mathrm{lo}} \le c_e < c_{\mathrm{hi}}$) requests a second view from the nearest neighbour at a lateral offset of $\pm b$ ($b = 15\,\mathrm{m}$) perpendicular to the observer–target direction, admitted only if the flank point keeps that neighbour's preserved links feasible. The flanking view enters the corroborative fusion below.

G. Corroborative multi-view fusion

A victim is observed by several UAVs from different directions. For each entry the ledger keeps the best confidence $c_{e,v}$ per viewing sector v (eight azimuth sectors around the entry), so repeated frames from one viewpoint are not counted twice. The textbook naive-Bayes rule $\operatorname{logit} c_e = \sum_v \operatorname{logit} c_{e,v}$ treats every low-confidence view as evidence *against* a victim.

Algorithm 1 SENTINEL agent i (every slot Δt)

sense: record look, add geolocated detections to the ledger (fused per viewpoint)
broadcast beacon $(x_i, \text{ledger}, \text{acks}, \text{heartbeat}, t_{\text{tether}})$; merge received beacons
if planning slot **then**
    update mode with (4); issue SA flank requests for own tentative entries
    if DELIVER and not connected **then**
        waypoint $\leftarrow x_0$
    else
        waypoint $\leftarrow \arg\max_{x \in \mathcal{D}_i} F_i(x)$ by PSO ▷ $S_i \equiv 0$ unless tethered
    end if
end if
$u_i \leftarrow$ step toward waypoint (rotated away from $\mathcal{O}$)
if tethered **then**
    $u_i \leftarrow \alpha^\star u_i$ with $\alpha^\star$ from (7)
end if
fly $x_i \leftarrow x_i + u_i$

Under occlusion this is wrong: a victim partly covered by a slab returns weak detections from the occluded side and strong ones from the open side, and a weak view from a blocked angle says little about absence. With many UAVs the many weak views outvote a clear view and the victim is never confirmed. We therefore fuse corroboratively,

$$\operatorname{logit} c_e = \max_v \operatorname{logit} c_{e,v} + \sum_{v \neq v^\star} \left[\operatorname{logit} c_{e,v}\right]_+, \qquad (9)$$

where $v^\star$ is the best view: the best view sets the confidence, other views can only add corroborating evidence, and a single view keeps its own confidence. Equation (9) is the component with the largest measured effect in dense swarms (Section VI-H); its cost is that two independent false alarms on the same spot corroborate each other, which we evaluate with deliberately adverse clutter (Section VI-G).

H. Complexity

Per planning step a UAV evaluates P particles for I iterations; each evaluation costs $O(k)$ for $k = |\mathcal{S}_i|$ outage look-ups plus an $O(1)$ map look-up (the footprint convolution is computed once per step, $O(M)$ for M map cells). The local MST costs $O(k_h^2 \log k_h)$ for the k_h neighbours heard. None of these terms depends on N; the measured per-agent compute time versus N is reported in Section VI.

V. IMPLEMENTATION

a) One algorithm library, two execution environments: All algorithms of Section IV (channel model, ledger, fusion, PSO, link preservation, delivery trigger) are implemented once in a ROS-independent Python library. The same library is executed by (i) a fast multi-agent simulator used for the statistical evaluation and (ii) ROS 2 Humble nodes that fly PX4 multicopters in Gazebo Harmonic, so the results of both environments refer to identical decision code.



Fig. 1. Generated 100 m disaster area in Gazebo Harmonic (top view): tall building shells (no-fly, link-obstructing) with their shadows, rubble blocks and concrete slabs. Victims and the six UAVs (launch point, bottom right) are too small to see at this scale; see Figs. 2 and 4.

b) Flight stack: Each UAV is a PX4 x500 quadrotor with a nadir thermal camera (640×512, 50° field of view, 10 Hz, radiometric 16-bit output converted to white-hot 8-bit on board). PX4 runs as software-in-the-loop; the Micro XRCE-DDS agent bridges its uORB topics to ROS 2. Per UAV, four nodes run on the companion computer (Fig. 3): `flight_commander` (offboard position control and world-frame pose), `thermal_vision` (YOLO26n detection and geolocation of every box into the world frame from the camera model, pose and heading), `emps_agent` (the SENTINEL agent) and the camera bridge. There is no central coordinator.

c) Radio emulation and failures: All inter-UAV beacons pass through a `comm_emulator` node that delivers each packet only if its sampled SINR, computed with the channel model of Section III and the building geometry of the world, exceeds the decoding threshold; it also hosts the responder radio. A `chaos_monkey` node injects crashes (force-disarm), radio cuts and brownouts on a schedule, and a `metrics_logger` computes the metrics of Section VI from ground truth. A beacon carries the sender's position, ledger, acknowledgements and heartbeat; it averages 4.1 kB (20 kB when the search map is included every fourth beacon).

d) Disaster scenes: Worlds are generated procedurally from a seed: tall building shells (no-fly, link-obstructing), low rubble that occludes victims, warm debris (316 K) that causes false alarms, and Rescue Randy mannequins with a 310 K body heat signature, partly or fully covered by concrete slabs (Figs. 1–2). The same seed produces the same scene in Gazebo and in the fast simulator, which is what makes the cross-check of Section VI-J possible.

e) Perception in the loop: Fig. 4 shows YOLO26n running on the Gazebo thermal feed during flight. Because the detector is trained on real thermal imagery, the rendered scenes

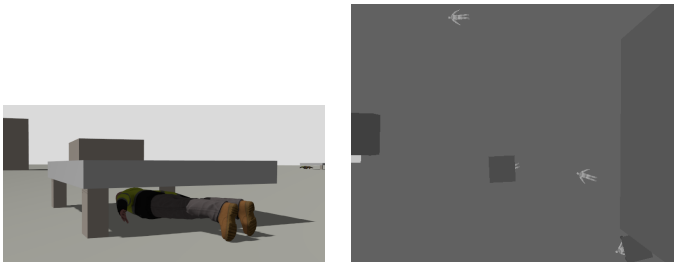


Fig. 2. Left: a victim partly covered by a concrete slab on supports; the torso is hidden from above and the legs are exposed. Right: white-hot thermal frame from UAV 0 at 30 m: two victims in the open, two under slabs with only the legs visible, and warm debris (bright square) beside a rubble block, a source of false alarms.

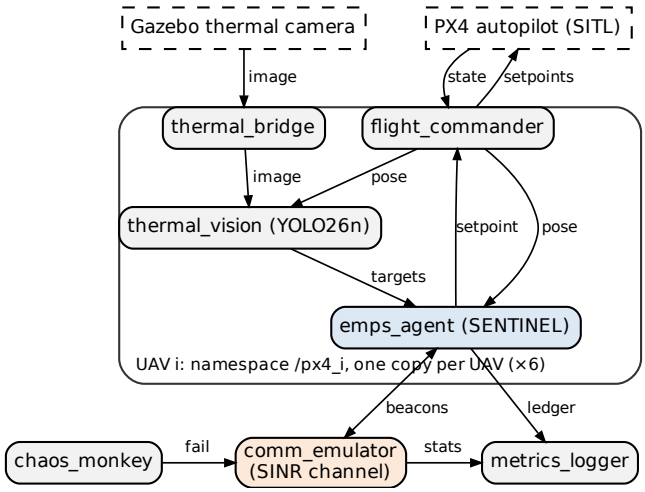


Fig. 3. ROS 2 node graph of the six-UAV demonstration, transcribed from the running system. The four nodes of one UAV are shown; each UAV runs its own copy. UAVs exchange information only as beacons through the radio emulator; there is no central coordinator.

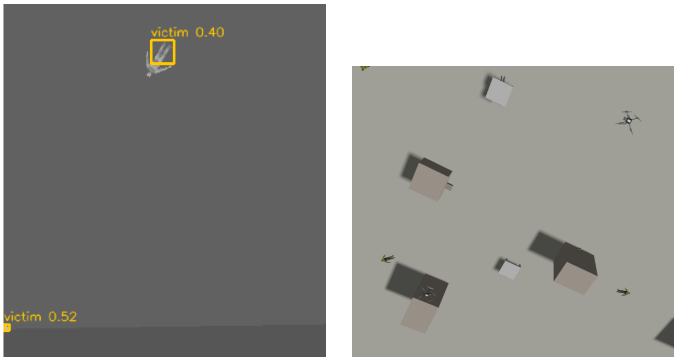


Fig. 4. Left: YOLO26n on the Gazebo thermal feed. The rendered mannequin is detected only in the SA band (0.40, orange) and a false box (0.52) appears on a building edge: the rendering gap to real thermal imagery. Right: two of the six UAVs searching at 30 m above victims, slabs and rubble.

TABLE III
DEFAULT PARAMETERS (FAST SIMULATOR AND SITL). †: TUNED (SECTION VI-A).

Group	Values
Kinematics	$\Delta t = 0.5$ s, $v = 5$ m/s, $h = 30$ m, $r_f = 12$ m, look every 1 s
Planning	every 1 s, horizon $T_p = 4$ s, PSO $P = 24$, $I = 12$, $\chi = 0.7298$, $c_{1,2} = 2.05$
Fitness	$w_c = w_d = 1$, $s = 8$ m, $w_t = 0.05$, $w_g = 0.066^{\dagger}$, $b_s = 1.03^{\dagger}$, $\ell_g = 60$ m
Spring	$\lambda_0 = 1.20^{\dagger}$, $\beta = 1.09^{\dagger}$, $\varepsilon = 0.05$
Delivery	$T_h = 116\,\mathrm{s}^{\dagger}$, orphan age $3T_h$, $T_{\mathrm{hb}} = 4$ s, $T_{\mathrm{nb}} = 3$ s, $T_{\mathrm{rel}} = 2$ s
Ledger, SA	gate 5 m, 8 sectors, $c_{\mathrm{lo}} = 0.40$, $c_{\mathrm{hi}} = 0.70$, $b = 15$ m
Detector	$c_{\max} = 0.92$, $\gamma = 1.5$, $r_{\mathrm{ref}} = 35$ m, $r_s = 60$ m, $\sigma_c = 0.08$, $\kappa_o = 0.8$
False alarms	0.02 per look, confidence $\mathcal{N}(0.22, 0.08^2)$
Objective	$\theta = 300$ s in (3)

produce false boxes on rubble and building edges and missed mannequins; we therefore evaluate detection accuracy on HIT-UAV (Section VI-B), and the cross-check of Section VI-J uses the detector model of the fast simulator, so that both environments see the same perception.

f) Reproducibility: A single command starts the complete Gazebo/PX4 demonstration; every experiment is defined by a configuration file, runs deterministically from its seed, and the tables and figures in this paper are generated by script from the stored results. The source code, scenes, configurations and raw results are released with the paper.

VI. EVALUATION

A. Methodology

a) Simulators: All algorithms are implemented once, in a ROS-independent library that is executed both by a fast multi-agent simulator and by the ROS 2 nodes running against PX4 SITL in Gazebo (Section V). The fast simulator advances all UAVs with a 0.5 s control step, samples every broadcast through the channel model of Section III (path loss, building obstructions, correlated shadowing, Rician/Rayleigh fading, collision interference) and produces detections with the sensing model of Section III. Each run starts from a procedurally generated disaster area (tall buildings that are no-fly zones and obstruct links, low rubble that occludes victims, warm debris that causes false alarms, 25 victims of which one is fully buried) and lasts 600 s. Unless stated otherwise the area is 400 m×400 m, $N = 8$, the channel is urban-rubble, and UAVs fail as a Poisson process with rate $0.01\,\mathrm{s}^{-1}$ per swarm. Table III lists the remaining parameters.

b) Metrics: The primary metric is the share of all victims whose report reaches the responder within 60 s of their first confirmation (*reported* ≤ 60 *s*); its denominator is the number of victims in the area, so a method cannot score well by detecting few. We also report the share of victims reported by the end of the mission, the 90th-percentile report latency, coverage of the reachable area, information loss (victims confirmed by some UAV whose every holder failed before the report reached the responder), false confirmed entries, and the per-agent planning time.

c) Baselines: B0: constriction PSO with the same exploration layer and no communication term; B1: the earlier SENTINEL design (nearest-neighbour distance spring, $R = 50$ m); B2: communication-aware PSO with a soft outage penalty on all neighbours; B3: algebraic-connectivity gradient control [2] given the *exact* λ_2 and Fiedler vector of the swarm at every

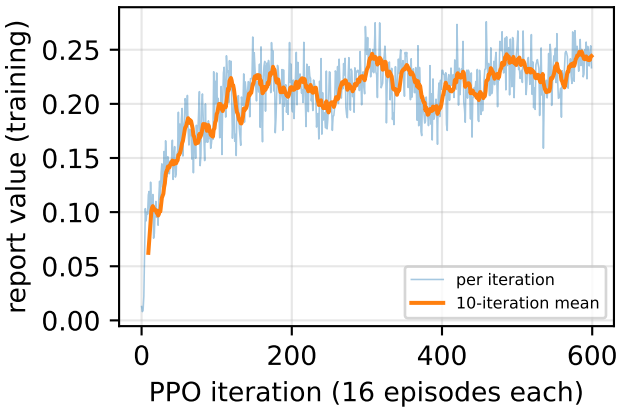


Fig. 5. Training curve of the MAPPO baseline (B5): time-discounted report value per PPO iteration and its 10-iteration mean. The return plateaus after about 150 iterations.

step (an oracle that favours B3); B4: edge-tension potential with hysteresis [4]; B5: MAPPO [17] with a parameter-shared actor on local observations and a centralised critic; B6: boustrophedon sweep of static sectors. All methods share sensing, gossip, the ledger and the channel. The baselines keep the best single view of each victim and do not request second views, as in their published designs; SENTINEL's fusion rule and flanking are therefore part of the comparison, and Section VI-H measures their share.

d) Fair tuning and statistics: Every method's free parameters were tuned with the same budget (Optuna TPE, 30 trials, each scored by the report value J of (3) on six runs in an urban $N = 8$ and a severe $N = 4$ scenario) on seeds 1000–1002, which are never used for evaluation. J rewards early reports over the whole mission and is therefore not identical to the 60 s metric; Section VII discusses the consequence for T_h. MAPPO was trained for 600 PPO iterations of 16 episodes each (9 600 missions) on separate seeds and selected on the same tuning seeds; its training return plateaus after about 150 iterations (Fig. 5), so the reported policy is not limited by training length. Each configuration is evaluated on seeds 0–29; all methods see the same area, victims and failure schedule for a given seed (common random numbers). We report means with 95 % bootstrap confidence intervals and test SENTINEL against every other method with paired Wilcoxon signed-rank tests, Holm-corrected within each scenario (marked * in the tables for $p < 0.05$). In total the evaluation comprises about 13 000 simulated missions.

B. Onboard perception

Table IV compares YOLO26n with YOLOv8n trained under an identical protocol on HIT-UAV [1]. The two detectors reach the same mAP50 (0.948 vs. 0.947); YOLO26n is more precise (0.932 vs. 0.909) and better localised (mAP50-95 0.533 vs. 0.520) with slightly lower recall, and it needs 19 % fewer parameters and 31 % fewer operations. On a desktop GPU the two are equally fast (about 1.8 ms end-to-end with TensorRT FP16); we therefore attribute YOLO26n's suitability for low-power hardware to its lower compute, not to measured speed, and we do not report embedded latency, which was not measured. The accuracy values are the validation metrics recorded by the training runs.

C. Communication conditions

Table V varies the channel from open terrain to the severe profile. SENTINEL reports the largest share of victims within 60 s in every profile and is significantly better than all seven baselines in the open, suburban and urban settings (e.g. 0.575 vs. 0.475 for the best baseline, B3, in urban rubble). In the severe channel it remains ahead (0.433 vs. 0.377) but the difference to the best baseline is not significant. Its total share of reported victims is 0.56–0.58 across profiles against 0.46–0.50 for the best baseline of each profile. In the severe profile SENTINEL's 90th-percentile latency (99 s) is longer than B0's (69 s): it delivers more of the reports that would otherwise never arrive, and some of these arrive late.

D. Failures

Tables VI and VII apply six failure settings: no failures, a single crash, 30 % of the swarm crashing simultaneously, three cascading crashes, Poisson crashes at twice the default rate, and a gradual transmit-power brownout. In the urban channel SENTINEL is significantly better than every baseline in every failure setting, by 23–30 % over the best baseline. In the severe channel it is ahead of or tied with the best baseline (often the lawnmower sweep), with no significant loss. Information loss, i.e. confirmed victims whose every holder failed before reporting, is small for *all* methods: at most 0.9 % averaged over all failure settings and at most 8.5 % under high-rate Poisson crashes in the severe channel (3.2 % for SENTINEL). Gossip replication between UAVs that meet during the search protects most information, so we make no claim that SENTINEL reduces loss compared with the baselines.

E. Scalability

Tables VIII and IX vary the swarm from 3 to 50 UAVs. On the 400 m area SENTINEL is significantly better than all baselines at every swarm size, from 0.203 vs. 0.153 with three UAVs to 0.788 vs. 0.725 with fifty. On the smaller 200 m area, where contacts are frequent, it has the highest mean at every swarm size but is significantly better than all baselines only for $N = 50$ (0.809 vs. 0.779). In dense swarms MAPPO (B5) becomes the strongest baseline (0.691–0.779 for $N \geq 10$ on the 200 m area), while it remains well behind in sparse settings. The per-agent planning time grows with the number of neighbours a UAV hears, not with N directly: from 4.0 to 11.1 ms on the dense 200 m area and from 5.1 to 9.5 ms on the 400 m area as N grows from 3 to 50 (Fig. 6), far below the 1 s planning period. The all-neighbour communication-aware PSO (B2) grows faster, to 20.6 ms.

F. Moving obstacles

With 5 or 15 tall debris objects drifting through the area, blocking links and hiding victims (Table X), SENTINEL

TABLE IV
THERMAL DETECTORS TRAINED WITH THE SAME PROTOCOL (HIT-UAV, 100 EPOCHS, ADAMW, 640 PX, SEED 0). ACCURACY: VALIDATION METRICS RECORDED BY THE TRAINING RUN. LATENCY: END-TO-END P50 ON AN RTX 3060 (DESKTOP), PYTORCH FP32 AND TENSORRT FP16.

Model	Params [M]	GFLOPs	mAP50	mAP50-95	P	R	PyTorch [ms]	TensorRT [ms]
YOLO26n	2.53	6.0	0.948	0.533	0.932	0.900	3.72	1.81
YOLOv8n	3.14	8.7	0.947	0.520	0.909	0.913	3.32	1.73

TABLE V
VICTIMS REPORTED WITHIN 60 S PER CHANNEL PROFILE (N=8, 400 M, POISSON FAILURES). MEAN OVER 30 SEEDS; BEST PER COLUMN IN BOLD; *: DIFFERS FROM SENTINEL (PAIRED WILCOXON, HOLM-CORRECTED, $p < 0.05$).

Method	open	severe	suburban	urban
SENTINEL	**0.577**	**0.433**	**0.560**	**0.575**
B0 PSO	0.473*	0.377	0.456*	0.453*
B1 Earlier design	0.457*	0.373	0.427*	0.409*
B2 Comm-PSO	0.472*	0.336*	0.371*	0.468*
B3 λ_2-grad.	0.477*	0.353	0.453*	0.475*
B4 Potential	0.501*	0.300*	0.099*	0.127*
B5 MAPPO	0.405*	0.333*	0.421*	0.396*
B6 Lawnmower	0.424*	0.373	0.423*	0.421*

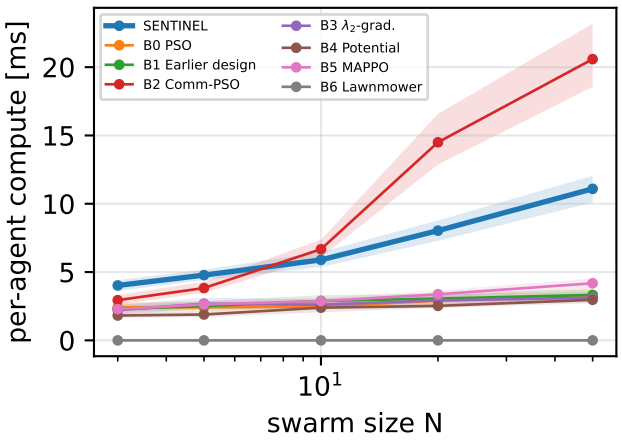


Fig. 6. Per-agent planning time vs. swarm size on the 200 m area (mean and 95 % CI over 30 seeds).

keeps reporting 0.58 of the victims within 60 s and remains significantly better than all baselines (best baseline 0.48–0.51).

G. Detection uncertainty

We perturbed the detector model around its default: confidence noise $\sigma \in \{0.04, 0.08, 0.15, 0.25\}$, a weaker detector ($c_{\max} \in \{0.8, 0.7\}$) and more frequent false alarms. SENTINEL stays significantly ahead of the two baselines run in this study, B0 and B6, in all eight settings (0.45–0.81 vs. at most 0.59). On-time reporting rises with σ for every method (SENTINEL 0.52 at $\sigma = 0.04$, 0.81 at $\sigma = 0.25$): confirmation requires $c \geq c_{\text{hi}}$ and the noise is symmetric, so a larger σ lets partly occluded victims, whose mean confidence lies below c_{hi}, cross the threshold more often, while false alarms keep their low confidences. The adverse case below removes this asymmetry. Because the default false-alarm confidences rarely exceed the confirmation threshold, we also ran a deliberately adverse case (Table XI): hot clutter that scores up to a mean of 0.55, the confidence noise applied to false alarms as well, and false-alarm rates of 0.1 and 0.3 per look (default 0.02). SENTINEL still reports 0.71–0.73 of the victims within 60 s against at most 0.50 (B0), with 0.18 false confirmed entries per mission on average (B0: 0.29). Moving the SA band and the confirmation threshold over four settings from 0.3/0.6 to 0.5/0.8 (default 0.4/0.7, $\sigma = 0.15$) changes the on-time share between 0.64 and 0.74 without producing false confirmations.

H. What produces the gains

Table XII removes one component at a time and Table XIII removes them jointly, on the same scenarios and seeds. Two components have significant effects. First, multi-view fusion: removing it reduces on-time reports significantly (urban, $N = 8$: 0.519 vs. 0.575), and in the dense swarm of Table XIV the corroborative rule is decisive (0.809, against 0.656 with plain naive-Bayes fusion and 0.720 for B0). The flank helps only together with fusion: without fusion, also removing the flank raises the on-time share (0.543 vs. 0.519), because flank requests then cost search time without adding evidence. Second, the delivery trigger: never delivering ($T_h = \infty$) lowers the urban on-time share from 0.575 to 0.523, and Fig. 7 sweeps T_h. With eight UAVs a short hold time is best in both channels: $T_h = 15$ s raises the on-time share over never delivering from 0.523 to 0.599 in the urban channel and from 0.419 to 0.501 in the severe channel, where it also shortens the 90th-percentile latency from 74 s to 41 s at almost the same total share (0.53); with four UAVs the differences are not significant. The remaining components (outage-based instead of distance-based spring, LMST instead of nearest-neighbour preserved set, information-dependent tension, projection) do not change the on-time share significantly; they act only during the short delivery tethers, and we examine them as the safety layer below.

Fusion is not part of the baselines' published designs, so Table XIII also gives every baseline SENTINEL's fusion rule. This raises the best urban baseline from 0.475 to 0.507 (B3), but SENTINEL remains significantly better than every baseline with fusion in the urban channel, with four and with eight UAVs; in the severe channel it remains ahead of all of them (significantly of B3 and B4 with four UAVs and of B4 and B5 with eight). In the dense swarm (Table XIV) the picture differs: with fusion, every baseline matches SENTINEL (0.815–0.849 vs. 0.809, not significant) and the edge-potential baseline B4 exceeds it (0.865, significant). There, SENTINEL's advantage

TABLE VI
VICTIMS REPORTED WITHIN 60 S UNDER FAILURE MODELS (URBAN-RUBBLE CHANNEL, N=8, 400 M). MEAN OVER 30 SEEDS; BEST PER COLUMN IN BOLD; *: DIFFERS FROM SENTINEL (PAIRED WILCOXON, HOLM-CORRECTED, $p < 0.05$).

Method	brownout	cascading	none	Poisson	30	
SENTINEL	**0.687**	**0.643**	**0.691**	**0.459**	**0.675**	**0.689**
B0 PSO	0.539*	0.509*	0.537*	0.344*	0.513*	0.523*
B1 Earlier design	0.513*	0.495*	0.501*	0.356*	0.499*	0.501*
B2 Comm-PSO	0.555*	0.524*	0.560*	0.340*	0.521*	0.545*
B3 λ_2-grad.	0.532*	0.489*	0.533*	0.339*	0.505*	0.523*
B4 Potential	0.183*	0.161*	0.193*	0.085*	0.144*	0.160*
B5 MAPPO	0.489*	0.457*	0.499*	0.328*	0.453*	0.473*
B6 Lawnmower	0.521*	0.488*	0.521*	0.345*	0.485*	0.511*

TABLE VII
VICTIMS REPORTED WITHIN 60 S UNDER FAILURE MODELS (SEVERE CHANNEL, N=8, 400 M). MEAN OVER 30 SEEDS; BEST PER COLUMN IN BOLD; *: DIFFERS FROM SENTINEL (PAIRED WILCOXON, HOLM-CORRECTED, $p < 0.05$).

Method	brownout	cascading	none	Poisson	30	
SENTINEL	**0.517**	0.452	**0.529**	**0.340**	**0.496**	**0.512**
B0 PSO	0.444	0.403	0.456	0.291	0.417	0.440
B1 Earlier design	0.448	0.423	0.469	0.295	0.440	0.452
B2 Comm-PSO	0.427*	0.385	0.439*	0.259*	0.400*	0.428
B3 λ_2-grad.	0.380*	0.363*	0.409*	0.249*	0.392*	0.409*
B4 Potential	0.359*	0.309*	0.396*	0.200*	0.359*	0.361*
B5 MAPPO	0.431*	0.383	0.445*	0.267*	0.412*	0.440*
B6 Lawnmower	0.499	**0.453**	0.508	0.287	0.468	0.495

TABLE VIII
VICTIMS REPORTED WITHIN 60 S VS SWARM SIZE N (400 M MAP, URBAN, POISSON FAILURES). MEAN OVER 30 SEEDS; BEST PER COLUMN IN BOLD; *: DIFFERS FROM SENTINEL (PAIRED WILCOXON, HOLM-CORRECTED, $p < 0.05$).

Method	N=3	N=5	N=10	N=20	N=50
SENTINEL	**0.203**	**0.389**	**0.667**	**0.751**	**0.788**
B0 PSO	0.153*	0.285*	0.535*	0.619*	0.656*
B1 Earlier design	0.109*	0.232*	0.499*	0.647*	0.707*
B2 Comm-PSO	0.147*	0.313*	0.535*	0.641*	0.685*
B3 λ_2-grad.	0.136*	0.273*	0.529*	0.636*	0.665*
B4 Potential	0.017*	0.027*	0.240*	0.485*	0.712*
B5 MAPPO	0.153*	0.273*	0.503*	0.628*	0.725*
B6 Lawnmower	0.151*	0.265*	0.444*	0.493*	0.619*

TABLE IX
VICTIMS REPORTED WITHIN 60 S VS SWARM SIZE N (200 M MAP, URBAN, POISSON FAILURES). MEAN OVER 30 SEEDS; BEST PER COLUMN IN BOLD; *: DIFFERS FROM SENTINEL (PAIRED WILCOXON, HOLM-CORRECTED, $p < 0.05$).

Method	N=3	N=5	N=10	N=20	N=50
SENTINEL	**0.475**	**0.620**	**0.720**	**0.743**	**0.809**
B0 PSO	0.457	0.597	0.660*	0.697*	0.720*
B1 Earlier design	0.360*	0.565	0.652*	0.692*	0.724*
B2 Comm-PSO	0.459	0.579	0.647*	0.725	0.732*
B3 λ_2-grad.	0.345*	0.560	0.664*	0.695*	0.727*
B4 Potential	0.044*	0.112*	0.471*	0.656*	0.735*
B5 MAPPO	0.379	0.559	0.691	0.729	0.779*
B6 Lawnmower	0.419	0.536*	0.541*	0.641*	0.703*

TABLE X
VICTIMS REPORTED WITHIN 60 S WITH MOVING OBSTACLES (URBAN, N=8). MEAN OVER 30 SEEDS; BEST PER COLUMN IN BOLD; *: DIFFERS FROM SENTINEL (PAIRED WILCOXON, HOLM-CORRECTED, $p < 0.05$).

Method	0 moving	5 moving	15 moving
SENTINEL	**0.575**	**0.585**	**0.579**
B0 PSO	0.453*	0.481*	0.460*
B1 Earlier design	0.409*	0.424*	0.447*
B2 Comm-PSO	0.468*	0.489*	0.509*
B3 λ_2-grad.	0.475*	0.479*	0.487*
B4 Potential	0.127*	0.132*	0.141*
B5 MAPPO	0.396*	0.449*	0.431*
B6 Lawnmower	0.421*	0.443*	0.425*

TABLE XI
VICTIMS REPORTED WITHIN 60 S WITH HOT-CLUTTER FALSE ALARMS (URBAN, N=8, σ=0.15). MEAN OVER 30 SEEDS; BEST PER COLUMN IN BOLD; *: DIFFERS FROM SENTINEL (PAIRED WILCOXON, HOLM-CORRECTED, $p < 0.05$).

Method	μ_{fp}=0.22, r=0.1	μ_{fp}=0.4, r=0.1	μ_{fp}=0.55, r=0.1	μ_{fp}=0.55, r=0.3
SENTINEL	**0.731**	**0.719**	**0.724**	**0.709**
B0 PSO	0.501*	0.491*	0.500*	0.493*
B6 Lawnmower	0.480*	0.480*	0.479*	0.481*
no SA	0.715	0.693	0.708	0.668
no fusion	0.672*	0.655*	0.676*	0.691

over the original baselines comes from the fusion rule alone, and its mobility gives no advantage when contacts are dense. Removing fusion, flank and delivery together leaves 0.524 in the urban channel, significantly below SENTINEL but still above every baseline with fusion. The rest of SENTINEL's margin over the original baselines is therefore not explained by these three components; the remaining differences are the tuned exploration aids and the heartbeat re-attachment, which we do not claim as contributions. Keeping the swarm connected at all times (SENTINEL-strict) lowers on-time reporting to 0.03–0.13.

I. Link preservation during delivery

Proposition 1 is a one-step guarantee for feasible steps. Over all experiments we counted the tethered, mutually preserved

TABLE XII
ABLATION: VICTIMS REPORTED WITHIN 60 S (400 M, POISSON FAILURES). MEAN OVER 30 SEEDS; BEST PER COLUMN IN BOLD; *: DIFFERS FROM SENTINEL (PAIRED WILCOXON, HOLM-CORRECTED, $p < 0.05$).

Method	severe, N=4	severe, N=8	urban, N=4	urban, N=8
SENTINEL	**0.212**	0.433	0.313	0.575
SENTINEL-strict	0.029*	0.061*	0.064*	0.131*
distance spring	0.200	0.433	0.307	0.561
NN set	**0.212**	0.433	0.313	0.573
no info. tension	0.211	0.433	0.312	0.572
no projection	0.200	0.435	0.308	0.545
no SA	0.189	**0.455**	**0.316**	**0.583**
no fusion	0.187	0.408	0.288	0.519*
naive-Bayes fusion	0.167	0.439	0.300	0.540

TABLE XIII
ATTRIBUTION: COMPONENTS REMOVED JOINTLY AND BASELINES WITH SENTINEL'S FUSION RULE; VICTIMS REPORTED WITHIN 60 S (400 M, POISSON FAILURES). MEAN OVER 30 SEEDS; BEST PER COLUMN IN BOLD; *: DIFFERS FROM SENTINEL (PAIRED WILCOXON, HOLM-CORRECTED, $p < 0.05$).

Method	severe, N=4	severe, N=8	urban, N=4	urban, N=8
SENTINEL	**0.212**	**0.433**	**0.313**	**0.575**
no fusion	0.187	0.408	0.288	0.519*
no SA, no fusion	0.168	0.425	0.307	0.543
no delivery (T_h=∞)	0.191	0.419	0.304	0.523*
no fusion, no delivery	0.175	0.393	0.289	0.496*
no SA, fusion, delivery	0.164	0.403	0.277	0.524*
B0 + fusion	0.169	0.392	0.227*	0.499*
B1 + fusion	0.172	0.417	0.203*	0.461*
B2 + fusion	0.153	0.353	0.228*	0.476*
B3 + fusion	0.127*	0.361	0.179*	0.507*
B4 + fusion	0.129*	0.300*	0.016*	0.119*
B5 + fusion	0.159	0.339*	0.213*	0.416*
B6 + fusion	0.159	0.396	0.224*	0.444*

links that were reliable before a step and unreliable after it: 93 violations in 1 349 703 link-steps (6.9×10^{-5}); with the swarm tethered at all times (SENTINEL-strict) the rate is 2.3×10^{-3}. To measure what the projection prevents, we also counted the same tethered links with the projection disabled (Table XIII scenarios and seeds): violations rise from 1 in 1 663 link-steps (6.0×10^{-4}, one of 120 runs) to 42 in 1 868 (2.2×10^{-2}, 21 runs; paired Wilcoxon $p < 10^{-4}$), a 37-fold reduction. The residual violations come from steps where the constraint was infeasible and the UAV hovered (Section IV-C). Because tethers are short (about 14 tethered link-steps per mission here), this protection does not measurably change the on-time share, which is why the layer does not appear among the sources of the gains. Information loss is highest without delivery in three of the four regimes of Fig. 7, the direction Proposition 3 predicts, but it does not decrease monotonically with T_h (at most 7.7 % in any cell), so the bound itself could not be tested quantitatively; it remains a design guarantee.

J. Cross-validation in PX4/Gazebo and the earlier scenario

To check that the fast simulator does not depend on idealised flight, we flew SENTINEL and B0 with three PX4 x500 multicopters in Gazebo Harmonic on the same three 100 m areas and seeds (300 s, crash of one UAV at 150 s), with the same agent code, tuned values, channel emulation and

TABLE XIV
DENSE SWARM (200 M, N=50): VICTIMS REPORTED WITHIN 60 S; ABLATIONS OF SENTINEL AND BASELINES WITH ITS FUSION RULE (SAME SEEDS). MEAN OVER 30 SEEDS; BEST IN BOLD; *: DIFFERS FROM SENTINEL (PAIRED WILCOXON, HOLM-CORRECTED, $p < 0.05$).

Method	50
SENTINEL	0.809
B0 PSO	0.720*
no SA	0.799
no fusion	0.740*
naive-Bayes fusion	0.656*
no SA, no fusion	0.719*
B0 + fusion	0.815
B1 + fusion	0.819
B2 + fusion	0.833
B3 + fusion	0.815
B4 + fusion	**0.865***
B5 + fusion	0.819
B6 + fusion	0.849

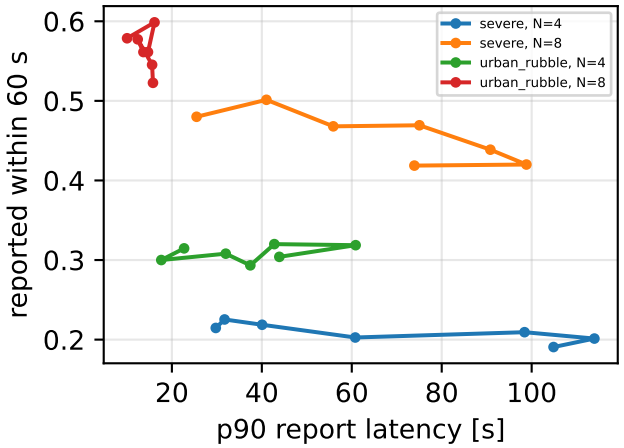


Fig. 7. Delivery trigger $T_h \in \{0, 15, 30, 60, 120, 240, \infty\}$: on-time share vs. 90th-percentile report latency for four regimes (each point: mean of 30 seeds).

detector model. The share of reported victims (SENTINEL 0.53 vs. 0.60 in the fast simulator; B0 0.60 vs. 0.62), the packet delivery ratio (0.79–0.81 vs. 0.80–0.83) and the connected fraction agree within 0.07; coverage of SENTINEL is higher in SITL (0.99 vs. 0.83). With three seeds this is a consistency check rather than a statistical comparison. In this 100 m area, which is comparable to the reliable range, B0 reports slightly more victims than SENTINEL in both environments; as in the 25 m scenario below, connectivity is rarely limiting there, so this check tests agreement between the environments, not the methods. Finally, re-running the scenario of the earlier SENTINEL design (25 m area, three UAVs, one crash) shows that it separates the methods only weakly: the area is far smaller than the radio range, every method stays connected throughout, and all of them report 63–71 % of the victims (SENTINEL 0.71, significantly above the earlier design with 0.63 but not above B0 with 0.67). The remaining victims are either buried or so close to another victim (4–5 m) that their reports merge (7–10 % of confirmations), which is a limitation of the association gate rather than of connectivity.

VII. Discussion and Limitations

a) Where the gains come from: The experiments support a narrower claim than "keeping the network alive". Connectivity in itself is not the bottleneck in the scenarios we studied: keeping the swarm connected at all times (SENTINEL-strict, B3, B4) limits the search and lowers on-time reporting, and gossip between UAVs that meet during the search already protects most information from failures. What raised timely reporting was deciding *when* a UAV should stop searching to deliver what it found, and combining views so that occlusion does not hide confirmed victims; these are the components with significant effects. Given the same fusion rule, the baselines remain significantly behind in the urban 400 m scenarios, but in a dense 50-UAV swarm they match SENTINEL and one exceeds it, so there the fusion rule, not the mobility, is what matters. Part of its margin in the urban default is not explained by these components (Section VI-H). The link-preservation layer (Proposition 1) did not change the on-time share measurably, but it reduced preserved-link violations 37-fold; its role is to make deliveries reliable when they happen.

b) Choice of T_h: T_h trades on-time reporting against total discovery. The value tuned on the report value J (116 s) is longer than the 60 s deadline. It still improves on-time reporting over never delivering (urban, $N = 8$: 0.577 at $T_h = 120$ s vs. 0.523), but a hold time of 15 s was better in both channels with eight UAVs (0.599 urban, 0.501 severe; Fig. 7). In practice T_h should follow the responders' reporting deadline; adapting it online to the observed contact rate is a natural extension.

c) Tuning sensitivity: An earlier configuration of SENTINEL that also tethered UAVs softly while exploring helped dense swarms but reduced coverage in the severe channel; the evaluated design does not use this tether. Results of that configuration are released with the others.

d) Limitations: (i) All results are from simulation: a fast multi-agent simulator and PX4 software-in-the-loop in Gazebo; no hardware flights were performed. (ii) The channel model is parameterised from published ranges and has not been calibrated against measurements in rubble. (iii) Detection accuracy is reported from the validation split recorded by the training runs; the detector transfers poorly to rendered thermal scenes, which is why the in-the-loop comparison uses the detector model; that model's parameters were chosen to reproduce the detector's qualitative behaviour and swept, not fitted to detector outputs. (iv) Victims closer than the association gate (about 5 m) can be merged into one report. (v) MAPPO was trained to a plateau (600 iterations) with one observation and reward design; other MARL formulations may perform better, in particular in dense swarms, where it is already the strongest baseline. (vi) The exposure bound (Proposition 3) could not be tested empirically because losses were rare at the failure rates studied. (vii) Latency was measured on a desktop GPU, not on embedded hardware.

VIII. Conclusion

We studied how a UAV swarm without infrastructure can get victim reports to responders in time. SENTINEL lets UAVs search freely, triggers delivery from the information each UAV has confirmed, fuses views so that occluded angles cannot veto a clear detection, and preserves the links of a delivery path for every feasible step (Proposition 1). In about 13 000 simulated missions against seven tuned baselines, it reported significantly more victims within 60 s than every baseline in most settings (urban, suburban and open channels, all failure settings in urban rubble, swarms of 3–50 UAVs on the 400 m area and moving obstacles), it stayed significantly ahead of the two baselines tested under detector uncertainty, in the severe channel it was ahead of or tied with the best baseline, and it was never significantly worse than any of the seven baselines. A PX4/Gazebo cross-check agreed with the fast simulator within 0.07. The ablation found significant effects for the fusion rule and the delivery trigger; in the urban 400 m scenarios SENTINEL remained significantly ahead when the baselines were given its fusion rule, whereas in a dense 50-UAV swarm the fusion rule alone explained its advantage. Continuous connectivity did not help, and the link-preservation layer reduced preserved-link violations 37-fold. Future work will adapt the delivery trigger to the observed contact rate, calibrate the channel model with measurements, and validate the system in hardware flights.

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